CSDA 5110
Spring 2024
Multiple Logistic Regression

1. Introduction

Objective:

- Generalize the logistic model to the case of more than one predictor/independent variable
- Strength of a modeling technique lies in its ability to model many variables.

Logistic Distribution

- Consider a collection of p independent variables denoted by the vector

$$\boldsymbol{x}^T = (x_1 , x_2, \ldots\ldots, x_p)$$

- P(Y=1|X=$\boldsymbol{x}$) = $\pi(\boldsymbol{x})$ must be greater than or equal to 0 and less than equal to 1.

$$E(Y|\boldsymbol{x}) = \pi(\boldsymbol{x}) = \frac{e^{\beta_0+\beta_1 x_1+\ldots\ldots+\beta_p x_p}}{1+e^{\beta_0+\beta_1 x_1+\ldots\ldots+\beta_p x_p}}$$

- Logit of the multiple logistic regression model is given by

$$\text{logit}\big(\pi(\boldsymbol{x})\big) = g(\boldsymbol{x}) = \log\left(\frac{\pi(\boldsymbol{x})}{1-\pi(\boldsymbol{x})}\right) = \beta_0 + \beta_1 x_1 + \ldots\ldots + \beta_p x_p$$

2. Fitting the Logistic Regression

- To fit the logistic regression to a given data requires estimation of the vector $\boldsymbol{\beta}^T = (\beta_0 , \beta_1, \ldots\ldots, \beta_p)$ of the unknown parameters

- Maximum likelihood method similar to the univariate case

- For logistic regression, there will be p+1 likelihood equations are:
  - $\sum_i^n [y_i - \pi(x_i)] = 0$ (1)
  - $\sum_i^n x_{ij}[y_i - \pi(\boldsymbol{x}_i)] = 0$, for *j = 1, 2, 3, …., p* (2)

- Will use software to obtain solutions

- $\hat{\pi}(\boldsymbol{x})$ is the maximum likelihood estimate of $\pi(\boldsymbol{x})$

- This quantity provides an estimate of the conditional probability that Y is equal to 1 given that $\mathbf{X} = \boldsymbol{x}$
- It represents the fitted or predicted value for the logistic regression model
- $\widehat{g(\boldsymbol{x})} = b_0 + b_1 x_1, \dots \dots . \beta_p x_p$

Variance- Covariance matrix

$$\widehat{Var}(\widehat{\boldsymbol{\beta}}) = (\boldsymbol{X}^T \boldsymbol{V} \boldsymbol{X})^{-1}$$

→ off diagonal Values are $Cov(\beta_i \beta_j)$ $i \neq j$

$$\widehat{SE}(\widehat{\boldsymbol{\beta}}) = \left(\widehat{Var}(\widehat{\boldsymbol{\beta}})\right)^{\frac{1}{2}}$$

$$\boldsymbol{X} = \begin{bmatrix} 1 & x_{11} & \cdot & \cdot & \cdot & x_{1p} \\ \cdot & x_{21} & x_{22} & & \cdot & x_{2p} \\ \cdot & \cdot & \cdot & & \cdot & \cdot \\ 1 & x_{n1} & & & & x_{np} \end{bmatrix}_{n \times p+1}$$

$$\boldsymbol{V} = \begin{bmatrix} \hat{\pi}_1(1-\hat{\pi}_1) & 0 & 0 & 0 \\ 0 & \hat{\pi}_2(1-\hat{\pi}_2) & 0 & 0 \\ 0 & 0 & \cdot & 0 \\ 0 & 0 & 0 & \hat{\pi}_p(1-\hat{\pi}_p) \end{bmatrix}$$

Test for $\boldsymbol{\beta}$: Wald's test statistic $\quad Z_j = \frac{b_j}{\widehat{SE}(b_j)}$

$H_0: \beta_j = 0$
$H_1: \beta_j \neq 0$

$\boxed{|z| = 1.96}$

Compare W to standard normal distribution

Multivariable analog of Wald's test

$$W = \widehat{\boldsymbol{\beta}}^T (\boldsymbol{XVX}) \widehat{\boldsymbol{\beta}}$$

Which is distributed chi-square with p+1 degrees of freedom.

Same as deviance method

$H_0: \beta_0 = \beta_1 = \beta_2 = \dots = \beta_p = 0$
$H_1:$ atleast one is $\neq 0$

Deviance:

In logistic regression the concept of the residual sum of squares is replaced by a concept known as the deviance.

The deviance associated with a given logistic regression model is based on comparing the maximized log-likelihood of the fitted model with the maximized log-likelihood under the so-called saturated model that has a parameter for each observation. In fact, the deviance is given by twice the difference between these maximized log-likelihoods.

H_0 : logistic regression model with p variables is appropriate
H_1 : logistic model is inappropriate

Under the null hypothesis, the deviance D^2 is approximately distributed as $\chi^2_{(p)}$, where n = the number of binomial samples, p = number of predictors in the model

$$G^2 = D(\textit{model without the variable}) - D(\textit{model with the variable})$$

$$G^2 = -2ln\left[\frac{(\textit{likelihood without the variable})}{(\textit{likelihood with the variable})}\right]$$

3. <u>Interpretation of the Fitted Logistic Regression Model</u>

Ability to draw practical inferences from the estimated coefficients in the model.

Interpretation involves two issues.

i. Determining the functional relationship between the dependent variable and the independent variable
ii. Appropriately defining the unit of change for the independent variable

- Proper interpretation of the coefficient in a logistic regression model depends on being able to place meaning on the difference between two logits
- OR = e^{β_1}; This simple relationship between the coefficient and the odds ration makes logistic regression a powerful analytical research tool
- It is the measure of association which approximates how much more likely (or unlikely) it is for the outcome to be present among those with x= a+1 than among those with x=a.

- Estimate of OR tends to have distribution that is skewed because the possible values range from 0 to infinity with the small null value equaling 1
- For large sample sizes, estimate of OR follows a normal distribution
- $\ln(\widehat{OR}) = \hat{\beta}_j$ follows a normal distribution even with small sample size

Dichotomous Independent Variable

Odds ratio is defined as ratio of the odds for x=1 to the odds for x=0.

$$OR = \frac{\frac{\pi(1)}{1-\pi(1)}}{\frac{\pi(0)}{1-\pi(0)}}$$

	x=1	x=0
y=1	$\pi(1) = \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}$	$\pi(0) = \frac{e^{\beta_0}}{1+e^{\beta_0}}$
y=0	$1-\pi(1) = \frac{}{1+e^{\beta_0+\beta_1}}$	$1-\pi(0) = \frac{}{1+e^{\beta_0}}$
	1	1

Example 2

x1 = Systolic blood pressure

x2 = A measure of cholesterol

x3 = A dummy variable (= 1 for patients with a family history)

x4 = A measure of obesity and

x5 = Age

```
Y = 1 when Heart Disease


View(heartdat)
head(heartdat)

## # A tibble: 6 × 7
##     Case    x1    x2    x3    x4    x5 HeartDisease
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>        <dbl>
## 1      1   160  5.73     1  25.3    52            1
## 2      2   144  4.41     0  28.9    63            1
## 3      3   118  3.48     1  29.1    46            0
## 4      4   170  6.41     1  32.0    58            1
## 5      5   134  3.5      1  26.0    49            1
## 6      6   132  6.47     1  30.8    45            0
```

Coefficients -Dichotomous Predictor (x3)

```
library(api2lm)
library(aod)
attach(heartdat)

crosstab <- table(HeartDisease, x3)
crosstab

##              x3
## HeartDisease   0   1
##             0 206  96
##             1  64  96

output_2 <- glm(HeartDisease~x3,family=binomial, heartdat)
summary(output_2)

##
## Call:
## glm(formula = HeartDisease ~ x3, family = binomial, data = heartdat)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.1774  -0.7356  -0.7356   1.1774   1.6968
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.1690     0.1431  -8.169 3.12e-16 ***
## x3            1.1690     0.2033   5.751 8.85e-09 ***


##confidence Interval
confint.default(output_2)

##                  2.5 %     97.5 %
## (Intercept) -1.4494758 -0.8885104
## x3           0.7706202  1.5673659

##odds ratio
exp(coef(output_2))

## (Intercept)          x3
##   0.3106796   3.2187500

## odds ratios and 95% CI
exp(cbind(OR = coef(output_2), confint(output_2)))

## Waiting for profiling to be done...

##                    OR    2.5 %    97.5 %
## (Intercept) 0.3106796 0.232958 0.4086384
## x3          3.2187500 2.167646 4.8124179
```

$e^{\hat{\beta}_3} = e^{1.169} = 3.2187$

odds of CHD among someone with family History increases by 3.22 times as compared to someone without family History

Coefficients -one continuous and one Dichotomous Predictor (x3)

```
library(api2lm)
library(aod)
library(vtable)

## Loading required package: kableExtra

## Warning in !is.null(rmarkdown::metadata$output) && rmarkdown::metadata$output
## %in% : 'length(x) = 2 > 1' in coercion to 'logical(1)'

summstat <- st(heartdat, group="x3")
summstat
```

Summary Statistics

x3	0			1		
Variable	**N**	**Mean**	**SD**	**N**	**Mean**	**SD**
Case	270	230	132	192	235	137
x1	270	137	21	192	140	20
x2	270	4.5	2	192	5.1	2.1
x4	270	26	4.2	192	27	4.2
x5	270	40	15	192	47	12
HeartDisease	270	0.24	0.43	192	0.5	0.5

```
output_1 <- glm(HeartDisease~x3+x1,family=binomial, heartdat)
summary(output_1)

##
## Call:
## glm(formula = HeartDisease ~ x3 + x1, family = binomial, data = heartdat)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.6198  -0.8707  -0.6624   1.1467   1.9659
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.741726   0.717515  -5.215 1.84e-07 ***
## x3           1.142013   0.206571   5.528 3.23e-08 ***
## x1           0.018545   0.005007   3.704 0.000212 ***
```

```r
##confidence Interval
confint.default(output_1)

##                    2.5 %      97.5 %
## (Intercept) -5.148030462 -2.33542206
## x3           0.737141483  1.54688362
## x1           0.008731314  0.02835874

##odds ratio
exp(coef(output_1))

## (Intercept)          x3          x1
##  0.02371313  3.13306748  1.01871805

## odds ratios and 95% CI
exp(cbind(OR = coef(output_1), confint(output_1)))

## Waiting for profiling to be done...

##                     OR     2.5 %     97.5 %
## (Intercept) 0.02371313 0.0056455 0.09472237
## x3          3.13306748 2.0961810 4.71505262
## x1          1.01871805 1.0088690 1.02893319
```

age adjusted odds ratio

3.13